

SHUBHAM VERMA

Machine Learning Engineer | Computer Vision | Generative AI

📍 Delhi, India



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GitHub



LinkedIn

EDUCATION

Vellore Institute of Technology

Bhopal, Madhya Pradesh

Bachelor of Technology in Computer Science Engineering with Specialization in Health Informatics 4 Years Program (2022-2026)

Overall GPA: 8.05/10.0

SKILLS

Languages	Python, C++, Java, Kotlin, SQL
Frameworks	Flask, FastAPI, Streamlit
ML/Data	Scikit-learn, Pandas, NumPy, OpenCV, PyTorch, Seaborn
AI/LLM Tools	LangChain, Pinecone, HuggingFace
Tools	Git, GitHub, MySQL, Android Studio
Concepts	Data Structures & Algorithms, REST APIs

PROJECTS

Vehicle Classification System (12 Classes) | PyTorch, EfficientNet, ONNX

[Git Link](#)

- Built a deep learning pipeline using EfficientNet-B0 achieving 85% validation accuracy and 0.85 F1-score on a 12-class dataset
- Improved dataset quality by removing 140+ noisy/duplicate samples (~5% of data), enhancing model generalization
- Addressed class imbalance across 12 categories using WeightedRandomSampler, improving minority class recall by ~12%
- Exported trained model to ONNX reducing inference overhead and enabling cross-platform deployment

Medical Chatbot (End-to-End RAG Pipeline) | Python, LangChain, Pinecone, Google Gemini

[Git Link](#)

- Architected a RAG-based medical chatbot reducing hallucinated responses by grounding outputs in verified medical literature
- Engineered a vector search pipeline using LangChain + Pinecone enabling sub-second (<1s) semantic retrieval
- Integrated Google Gemini Pro to generate context-aware responses with improved factual accuracy
- Served chatbot via web interface handling real-time query processing

AI Mental Health Assistant (SafeSpace) | LangGraph, FastAPI, Streamlit, Ollama

[Git Link](#)

- Built an AI agent-based mental health assistant integrating LLM reasoning with tool-based actions
- Designed a full-stack system with FastAPI backend and Streamlit frontend enabling real-time conversational interactions
- Implemented tool-calling workflows for emergency alerts and therapist recommendations based on user intent
- Achieved real-time conversational response latency of ~1-2 seconds using optimized LLM inference

EXPERIENCE

Android Development Intern | Ashva Wearable Technologies

Sep 2024 – Jan 2025

- Developed cross-platform Android application using Kotlin Multiplatform (KMP) supporting real-time BLE data streaming at 128 data points/sec
- Built low-latency UI using Jetpack Compose ensuring real-time visualization and processing of incoming sensor data
- Implemented on-device data transformation and computation, reducing dependency on external processing and improving responsiveness

Software Development Intern | Neuramics Lab

Feb 2026 – July 2026

- Reduced model inference latency by 2.5× by migrating workloads from CPU to GPU using CUDA acceleration
- Developed computer vision pipelines processing ~40 images/sec for masking and background correction
- Deployed optimized models on NVIDIA Jetson Orin Nano achieving real-time inference at ~40 FPS

CERTIFICATIONS

- Applied Machine learning in python (Coursera Course)
- Java Development on Oracle Cloud (Oracle)
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional (Oracle)
- Oracle Cloud Infrastructure 2025 Data Science Professional (Oracle)

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EXTRACURRICULAR ACTIVITIES

- Member Of Data Science Club
- Participated in “ECO - Hackathon” (March 2025)
- Participated in “L’Oreal Sustainability Challenge” (February 2025)
- Finalist at HackFinance - IIITD Fintech Hackathon (Student Division)

[certificate](#)

[certificate](#)